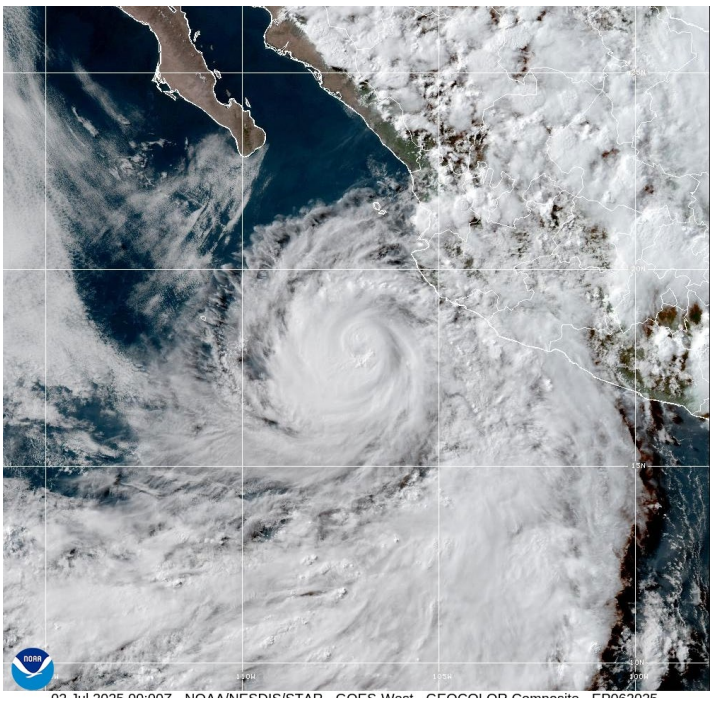
# NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

# HURRICANE FLOSSIE

## (EP062025)

## 29 June – 3 July 2025

Larry A. Kelly
National Hurricane Center
30 July 2025



**GOES-WEST GEOCOLOR SATELLITE IMAGE OF HURRICANE FLOSSIE AROUND THE TIME OF ITS PEAK INTENSITY AT 0000 UTC 2 JULY 2025, AS IT PASSED JUST OFFSHORE THE COAST OF SOUTHWESTERN MEXICO. IMAGE COURTESY NOAA/NESDIS/STAR.**

Flossie was a category 3 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that passed just offshore the coast of southwestern Mexico. High surf generated by the hurricane caused one direct fatality in Mexico.

# Hurricane Flossie

**29 June – 3 July 2025**

## SYNOPTIC HISTORY

Flossie developed from a tropical wave that crossed Central America on 25 June. The tropical wave then started to interact with the monsoon trough over the far eastern Pacific. The combined feature gradually lifted northward over the next few days, and showers and thunderstorms began to organize on 28 June. Satellite-derived wind data from ASCAT indicated that a well-defined low-level center had formed with sufficiently organized deep convection by 0600 UTC 29 June, marking the formation of a tropical depression about 235 n mi south of Acapulco, Mexico. The depression strengthened into Tropical Storm Flossie 12 hours later. The "best track" chart of Flossie's path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[1].

After Flossie formed, it moved west-northwestward to northwestward parallel to the southwestern coast of Mexico. The storm moved over warm ocean waters of 30°C and within a low wind shear environment favorable for strengthening, Flossie rapidly intensified from a 35-kt tropical storm at 0000 UTC 30 June to a 65-kt hurricane at 0000 UTC 1 July, becoming the third hurricane of the 2025 eastern Pacific basin hurricane season. The hurricane made its closest approach to the west-central coast of Mexico as it continued to intensify on 1 July, with far outer rain bands impacting coastal areas. However, sustained tropical-storm-force winds remained offshore. Flossie began moving farther away from Mexico late on 1 July toward a weakness in the subtropical ridge caused by a mid- to upper-level trough over California, and it reached its peak intensity of 105 kt at 0300 UTC 2 July.

Flossie continued west-northwestward during the next few days, crossing over cooler sea surface temperatures, and into a drier, more stable air mass. This environment induced rapid weakening beginning later on 2 July into 3 July. Flossie started to lose deep convection early on 3 July, and it became a post-tropical cyclone at 1200 UTC that day. The cyclone continued to weaken and moved west-northwest over the next couple of days before opening into a trough by 0000 UTC 6 July.

## METEOROLOGICAL STATISTICS

Observations in Flossie (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting

[1] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Flossie.

There were no ship or land-based reports of winds of tropical storm force associated with Flossie.

### *Winds and Pressure*

Flossie's estimated peak intensity of 105 kt at 0300 UTC 2 July is based on a blend of UW-CIMSS ADT, DMINT and SATCON objective intensity estimates in conjunction with a more well-defined eye on infrared satellite imagery. The TAFB Dvorak intensity estimate at 0600 UTC 2 July was T5.5/102 kt but noted the appearance of the eye had become more ragged in the last hour or so before the synoptic time.

The highest land-based wind report was at Aeropuerto Internacional Playa de Oro located in Manzanillo, Mexico (MMZO), that had a maximum sustained wind speed of 30 kt and gust to 40 kt at 0055 UTC 2 July within an outer rain band.

The estimated minimum central pressure of 958 mb is primarily based on the Knaff-Zehr-Courtney pressure-wind relationship. Pressure estimates based on subjective and objective satellite intensity estimates were a couple of millibars lower.

### *Rainfall and Flooding*

Rainfall from Flossie's outer rain bands caused some minor flooding across portions of southwestern Mexico, though there was already saturated ground with an earlier event prior to Flossie producing 1 to 3 inches (25 mm to 75 mm) on 29 June.

## CASUALTY AND DAMAGE STATISTICS

Flossie is responsible for at least 1 direct fatality[2] in Mexico. A young man entered the rough surf early in the morning of 30 June at Playa Olas Altas in the municipality of Manzanillo where he was swept away.

[2] Deaths occurring as a direct result of the forces of the tropical cyclone are referred to as "direct" deaths. These would include those persons who drowned in storm surge, rough seas, rip currents, and freshwater floods. Direct deaths also include casualties resulting from lightning and wind-related events (e.g., collapsing structures). Deaths occurring from such factors as heart attacks, house fires, electrocutions from downed power lines, vehicle accidents on wet roads, etc., are considered "indirect" deaths.

# FORECAST AND WARNING CRITIQUE

## *Genesis*

The possibility of Flossie's genesis was noted more than a week in advance, but the system took longer to develop than expected. Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each category. A low (<40%) chance of genesis during the next 7 days was first indicated in the TWO about ten days (234 hours) before Flossie formed, and the chances were raised to the medium (40–60%) category 204 h before formation. For the 2-day forecast period, the disturbance was given a low and medium chance of genesis 168 and 102 hours before formation, respectively. The 7-day and 2-day probabilities were raised to the high categories 186 h and 24 h before formation, respectively. Despite the lack of confidence on when Flossie would form, the forecasts for the cyclone's genesis location were very accurate and had a 100 percent hit rate, lying within every genesis area depicted in NHC's Graphical TWO (Fig. 4). When the system had a low and medium chance of development, the location of genesis occurred on the northwestern edge of the NHC genesis area, indicating the system formed later than originally anticipated.

## *Track*

A verification of NHC official track forecasts for Flossie is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at each time range. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The NHC's track forecast for Flossie demonstrated good performance. The simple consensus aids, specifically TVCE, TVCX, and TVDG, proved most accurate. These models predicted Flossie would remain slightly farther offshore of Mexico, as compared to the corrected-consensus HCCA and Google DeepMind models, particularly at the 60- and 72-hour forecast periods.

## *Intensity*

A verification of NHC official intensity forecasts for Flossie is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period, with the exception of the 12-hour forecast, which showed comparable errors. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. The NHC official intensity forecast for Flossie performed well, outperforming most available guidance. The corrected consensus HCCA model was the most accurate. In contrast, the hurricane regional models, in particular the HMON, struggled to correctly forecast Flossie's rapid intensification.

### *Wind Watches and Warnings*

Coastal wind watches and warnings associated with Flossie are given in Table 5.

## ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the Graphical TWO verification figure.

Table 1. Best track for Hurricane Flossie, 29 June – 3 July 2025.

| Date/Time (UTC) | Latitude (°N) | Longitude (°W) | Pressure (mb) | Wind Speed (kt) | Stage |
|---|---|---|---|---|---|
| 29 / 0600 | 13.0 | 99.1 | 1008 | 25 | tropical depression |
| 29 / 1200 | 13.1 | 99.6 | 1006 | 30 | " |
| 29 / 1800 | 13.4 | 100.1 | 1005 | 35 | tropical storm |
| 30 / 0000 | 14.0 | 100.7 | 1005 | 35 | " |
| 30 / 0600 | 14.6 | 101.4 | 1004 | 40 | " |
| 30 / 1200 | 15.3 | 102.1 | 999 | 45 | " |
| 30 / 1800 | 15.9 | 103.0 | 996 | 55 | " |
| 01 / 0000 | 16.4 | 103.9 | 988 | 65 | hurricane |
| 01 / 0600 | 16.8 | 104.7 | 984 | 70 | " |
| 01 / 1200 | 17.2 | 105.5 | 978 | 80 | " |
| 01 / 1800 | 17.6 | 106.3 | 966 | 95 | " |
| 02 / 0000 | 18.0 | 107.1 | 962 | 100 | " |
| 02 / 0300 | 18.2 | 107.5 | 958 | 105 | " |
| 02 / 0600 | 18.4 | 107.9 | 962 | 100 | " |
| 02 / 1200 | 18.8 | 108.7 | 967 | 95 | " |
| 02 / 1800 | 19.3 | 109.4 | 974 | 85 | " |
| 03 / 0000 | 19.6 | 109.9 | 983 | 70 | " |
| 03 / 0600 | 20.0 | 110.6 | 997 | 50 | tropical storm |
| 03 / 1200 | 20.4 | 111.3 | 1001 | 40 | low |
| 03 / 1800 | 20.9 | 112.0 | 1003 | 35 | " |

| Date/Time (UTC) | Latitude (°N) | Longitude (°W) | Pressure (mb) | Wind Speed (kt) | Stage |
|---|---|---|---|---|---|
| 04 / 0000 | 21.6 | 112.9 | 1006 | 30 | " |
| 04 / 0600 | 22.3 | 113.8 | 1007 | 30 | " |
| 04 / 1200 | 22.9 | 114.8 | 1007 | 30 | " |
| 04 / 1800 | 23.4 | 115.8 | 1008 | 25 | " |
| 05 / 0000 | 23.8 | 116.6 | 1010 | 20 | " |
| 05 / 0600 | 24.0 | 117.3 | 1011 | 15 | " |
| 05 / 1200 | 24.1 | 117.9 | 1013 | 15 | " |
| 05 / 1800 | 24.1 | 118.5 | 1015 | 15 | " |
| 06 / 0000 | | | | | dissipated |
| 02 / 0300 | 18.2 | 107.5 | 958 | 105 | maximum winds and minimum pressure |

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

| | Hours Before Genesis | |
|---|---|---|
| | **48-Hour Outlook** | **168-Hour Outlook** |
| Low (<40%) | 168 | 234 |
| Medium (40%-60%) | 102 | 204 |
| High (>60%) | 24 | 186 |

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Flossie, 29 June – 3 July 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

| | Forecast Period (h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 60 | 72 | 96 | 120 |
| OFCL | **12.7** | **16.6** | **16.2** | **17.9** | **31.3** | **48.4** | **66.8** | |
| OCD5 | 24.8 | 48.3 | 67.5 | 84.6 | 110.6 | 155.5 | 260.8 | |
| Forecasts | 15 | 13 | 11 | 9 | 7 | 5 | 1 | |
| OFCL (2020-24) | 22.4 | 33.6 | 43.5 | 53.2 | 65.1 | 79.4 | 108.6 | 135.2 |
| OCD5 (2020-24) | 38.2 | 75.3 | 117.2 | 161.0 | 206.0 | 252.0 | 340.7 | 422.7 |

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Flossie, 29 June – 3 July 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

| Model ID | Forecast Period (h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 60 | 72 | 96 | 120 |
| OFCL | 12.7 | 17.3 | 16.6 | 19.4 | 34.3 | 52.2 | | |
| OCD5 | 21.8 | 42.8 | 58.8 | 73.7 | 97.7 | 142.1 | | |
| GFSI | 12.9 | 22.4 | 33.1 | 47.4 | 53.1 | **47.2** | | |
| HWFI | 16.3 | 35.0 | 56.4 | 66.8 | 65.1 | **45.1** | | |
| HMNI | 15.4 | 25.6 | 36.0 | 47.0 | 58.6 | 57.8 | | |
| HFAI | 14.1 | 19.0 | 26.4 | 31.0 | 40.9 | **38.6** | | |
| HFBI | 14.7 | 22.0 | 28.5 | 41.5 | 58.4 | 70.4 | | |
| GDMI | 13.6 | 19.6 | 23.2 | 31.3 | 50.9 | 70.0 | | |
| EMXI | 18.4 | 24.1 | 25.7 | 23.0 | **30.5** | **34.0** | | |
| NVGI | 21.3 | 35.0 | 36.6 | 31.3 | 34.4 | **42.4** | | |
| CMCI | 20.5 | 36.6 | 44.8 | 44.7 | 60.5 | 108.2 | | |
| CTCI | 14.4 | 22.8 | 32.9 | 41.5 | 48.9 | 59.6 | | |
| TVCE | **11.0** | **14.1** | **16.5** | 20.9 | **27.6** | **30.8** | | |
| TVCX | **12.3** | **14.3** | **15.4** | 22.0 | **26.9** | **29.1** | | |
| GFEX | 14.9 | 18.5 | 19.8 | 27.9 | **33.0** | **31.2** | | |
| TVDG | **11.9** | **14.4** | **15.7** | 22.5 | **29.3** | **29.8** | | |
| HCCA | 13.9 | 19.2 | 22.9 | 26.2 | 45.9 | 71.2 | | |
| FSSE | 14.7 | 17.7 | **15.8** | **19.2** | 34.6 | **48.9** | | |

| Model ID | Forecast Period (h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 60 | 72 | 96 | 120 |
| AEMI | 13.2 | 19.6 | 29.8 | 41.3 | 48.9 | **41.3** | | |
| TABS | 34.7 | 82.8 | 126.8 | 169.7 | 215.6 | 259.7 | | |
| TABM | 30.2 | 70.2 | 102.1 | 132.6 | 174.9 | 224.8 | | |
| TABD | 26.4 | 61.3 | 94.5 | 125.9 | 171.3 | 240.5 | | |
| Forecasts | 14 | 12 | 10 | 8 | 6 | 4 | | |

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Flossie, 29 June – 3 July 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

| | Forecast Period (h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 60 | 72 | 96 | 120 |
| OFCL | 7.7 | **8.5** | **10.0** | **12.8** | **13.6** | **11.0** | **15.0** | |
| OCD5 | 10.8 | 18.2 | 22.2 | 25.3 | 30.9 | 24.2 | 8.0 | |
| Forecasts | 15 | 13 | 11 | 9 | 7 | 5 | 1 | |
| OFCL (2020-24) | 5.7 | 8.9 | 10.8 | 12.9 | 14.4 | 15.5 | 17.0 | 18.6 |
| OCD5 (2020-24) | 7.4 | 12.4 | 16.2 | 19.2 | 21.0 | 21.3 | 21.5 | 21.6 |

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Flossie, 29 June – 3 July 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

| Model ID | Forecast Period (h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 60 | 72 | 96 | 120 |
| OFCL | 7.9 | 8.8 | 10.0 | 13.1 | 12.5 | 10.0 | | |
| OCD5 | 11.2 | 19.6 | 23.5 | 26.4 | 28.5 | 18.2 | | |
| HWFI | 9.0 | 13.9 | 16.9 | 16.8 | 18.5 | 15.5 | | |
| HMNI | 8.6 | 16.2 | 25.3 | 28.5 | 28.3 | 21.8 | | |
| HFAI | 7.9 | 13.8 | 15.4 | 14.6 | 17.7 | 14.5 | | |
| HFBI | **7.1** | 12.8 | 16.1 | 16.6 | 17.2 | 15.5 | | |
| GDMI | **7.3** | 11.2 | 14.2 | 17.9 | 19.2 | 16.0 | | |
| DSHP | 8.4 | 12.8 | 15.4 | 15.2 | **9.8** | 12.8 | | |
| LGEM | 8.4 | 14.3 | 15.3 | 17.0 | **10.0** | **7.5** | | |
| ICON | 8.0 | 13.8 | 15.8 | 16.9 | 15.7 | 10.8 | | |
| IVCN | **7.5** | 13.4 | 15.4 | 15.0 | 13.8 | 11.0 | | |
| IVDR | **7.6** | 13.3 | 16.7 | 16.4 | 15.2 | 12.2 | | |
| CTCI | 8.2 | 16.2 | 17.4 | 14.6 | **12.2** | 15.8 | | |
| GFSI | 9.6 | 15.8 | 20.6 | 22.1 | 20.5 | 14.0 | | |
| EMXI | 9.9 | 17.9 | 24.6 | 30.9 | 30.7 | 26.2 | | |
| HCCA | **6.1** | 9.3 | 12.6 | **10.5** | 13.0 | **8.5** | | |
| FSSE | **6.5** | 11.2 | 15.5 | 16.6 | 16.5 | 11.0 | | |
| Forecasts | 14 | 12 | 10 | 8 | 6 | 4 | | |

Table 5. Watch and warning summary for Hurricane Flossie, 29 June – 3 July 2025.

| Date/Time (UTC) | Action | Location |
|---|---|---|
| **29/ 0900** | Tropical Storm Watch Issued | Southwest coast of Mexico from Zihuantanejo northward to Manzanillo |
| **29/ 1500** | Tropical Storm Watch Issued | Southwest coast of Mexico from Manzanillo northward to Cabo Corrientes |
| **30/ 0900** | Tropical Storm Warning issued | Southwest coast of Mexico from Punta San Telmo northward to Playa Perula |
| **01/ 2100** | Tropical Storm Watch discontinued | All of Mexico |
| **02/ 0900** | Tropical Storm Warning discontinued | All of Mexico |

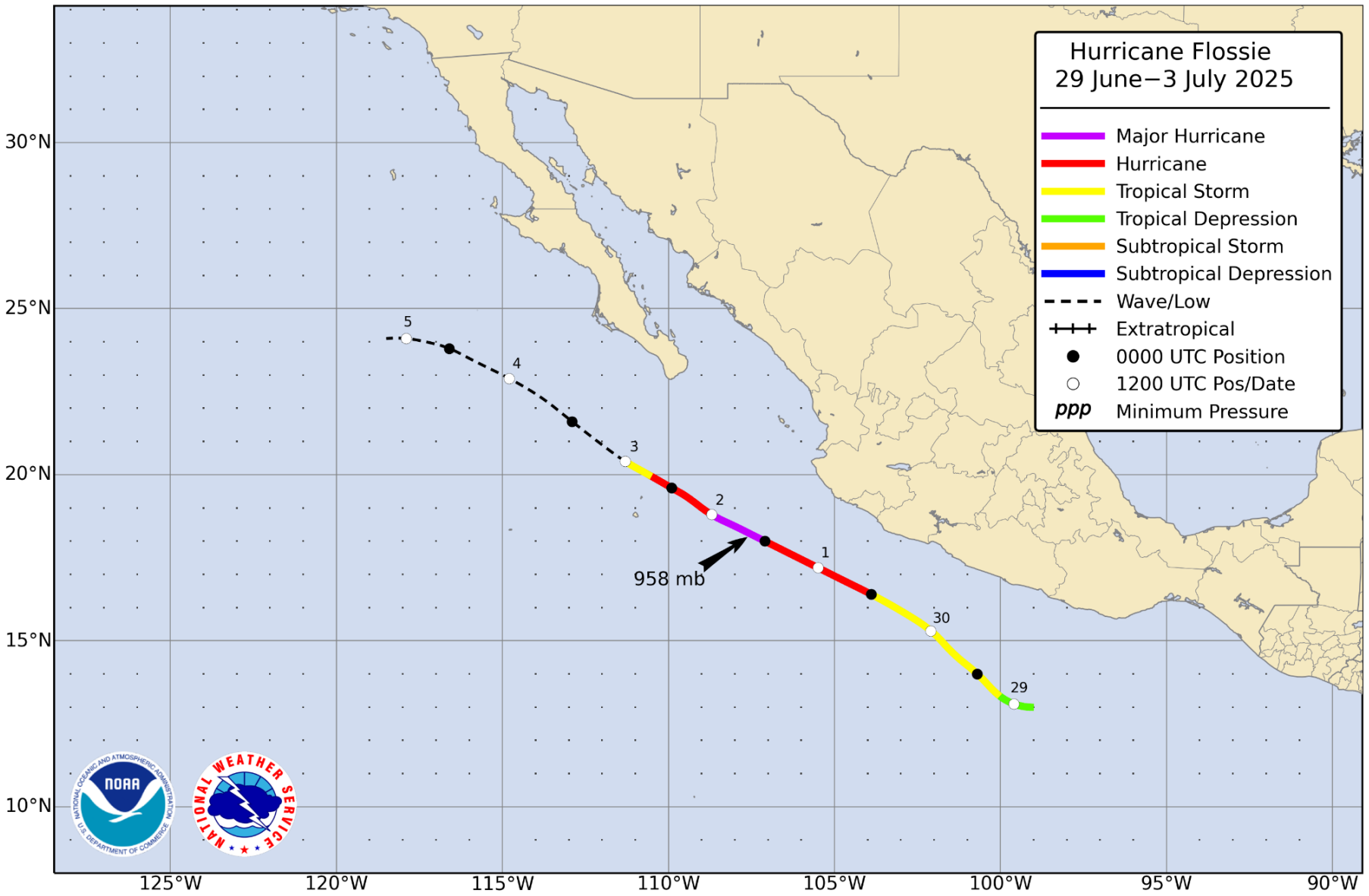



Figure 1. Best track positions for Hurricane Flossie, 29 June – 3 July 2025.

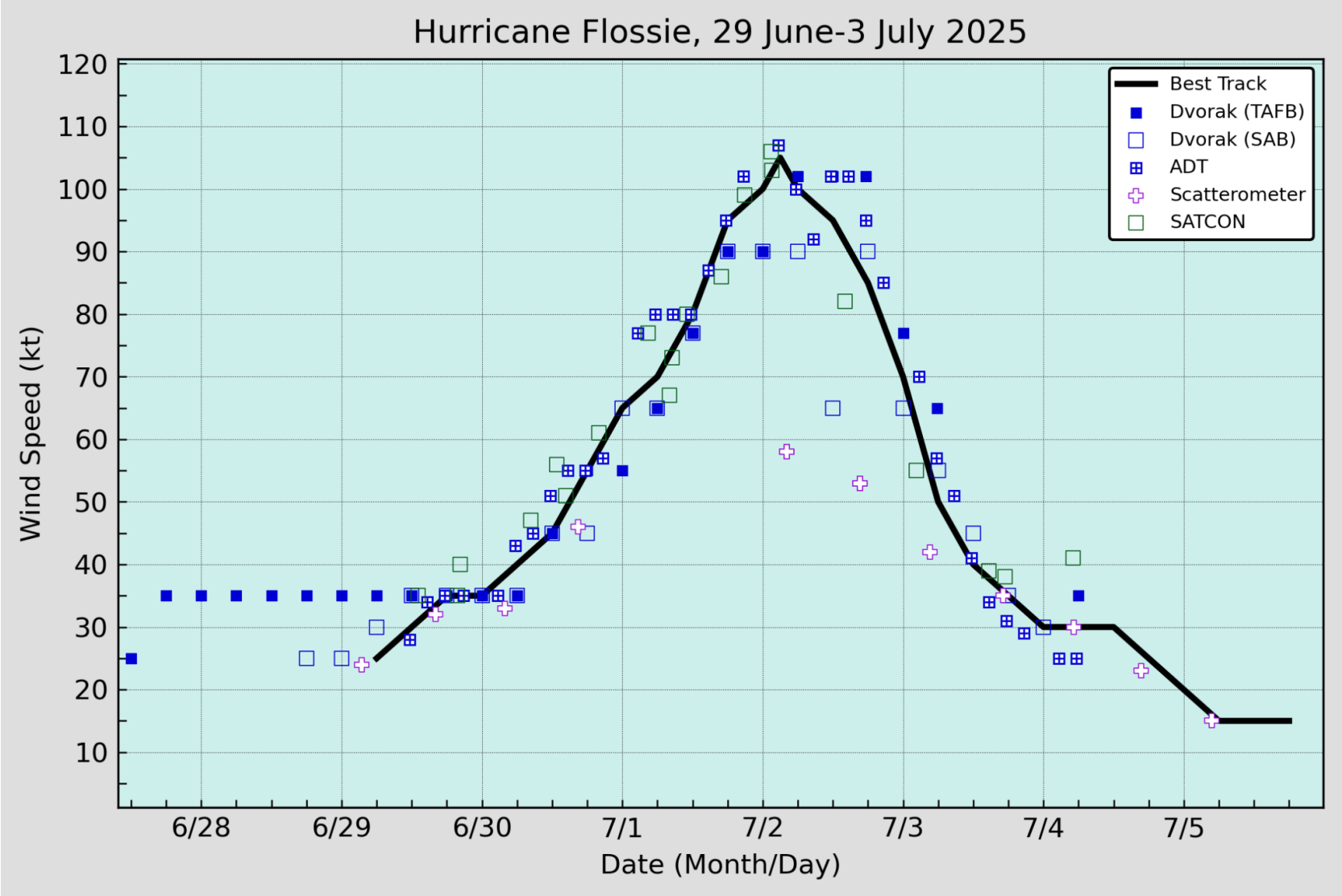



Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Flossie, 29 June – 3 July 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

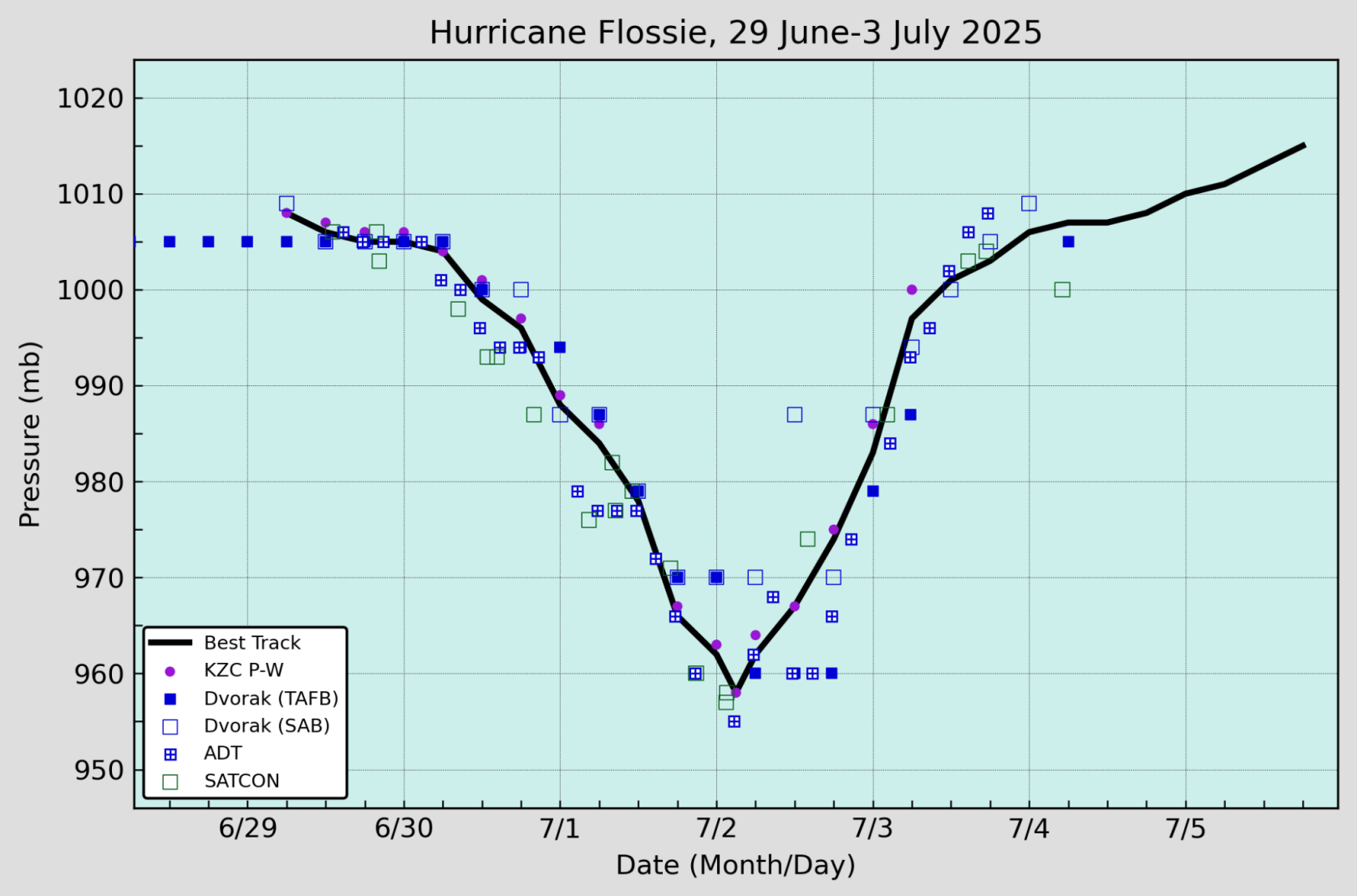



Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Flossie, 29 June – 3 July 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

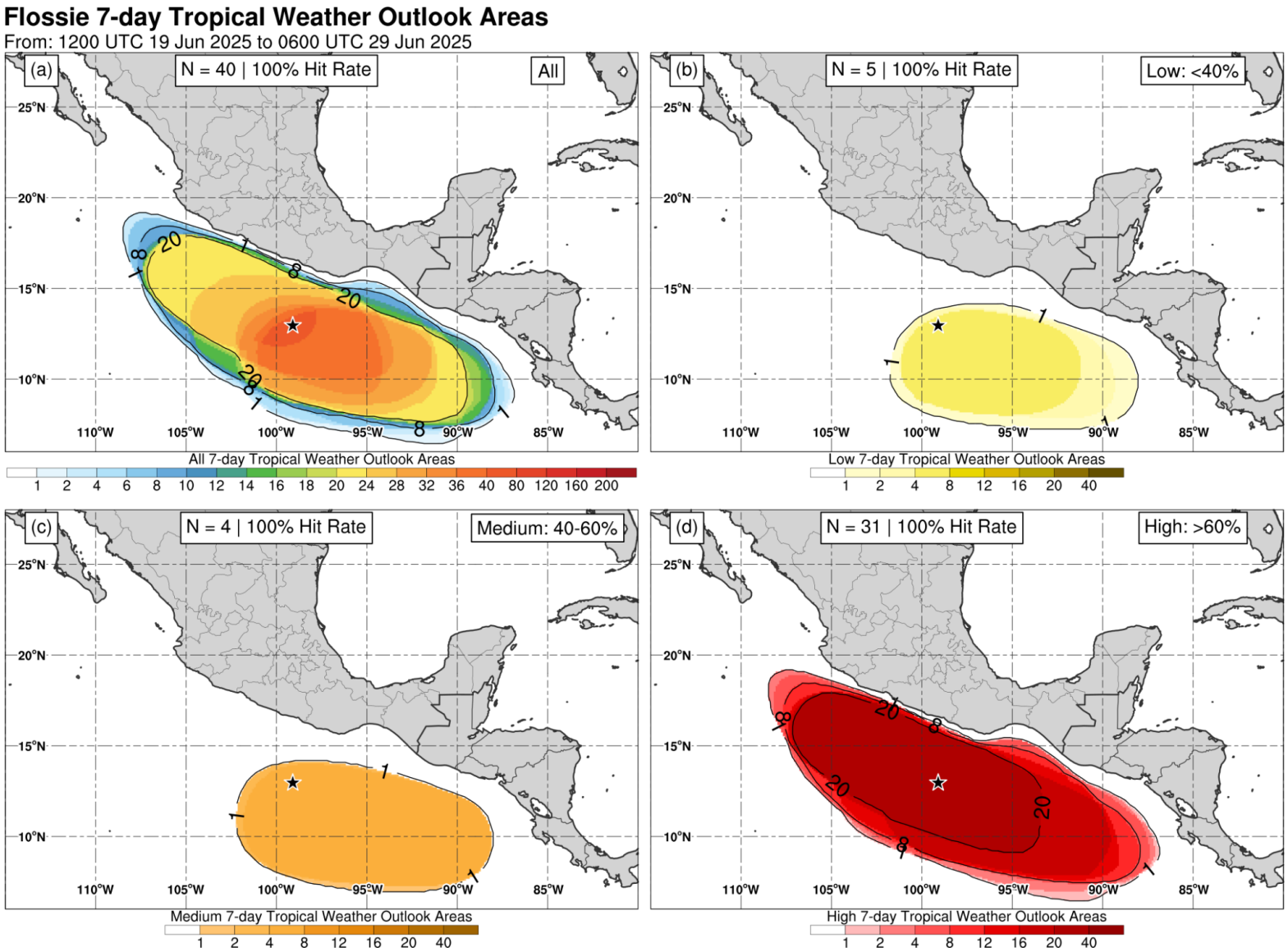



Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Hurricane Flossie for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.